Myosin is essential for the growth and development of gametophytes in *Ceratopteris richardii.*

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ABSTRACT

Myosin is a GTPase motor protein that is present in eukaryotes, playing a vital role in many cellular processes. In seed plants, there are two types of myosin that are found: Type VIII and XI. These myosins have been shown to regulate cytoplasmic streaming and organelle trafficking. Little is known about the role of myosin in seedless plants. This study identifies and characterizes an orthologue of a Type XI myosin in the seedless plant *Ceratopteris richardii*.

When sterilized spores were treated with concentration 10 µM, 20 µM and 50 µM of 2,3-butanedione monoxime (BDM), a chemical inhibitor of the myosin, the rate of germination decreased as the concentration of BDM increased. If BDM is added to spores already germinated, 10 µM of BDM is sufficient to reduce cell division and rhizoid elongation; decreasing the overall area of the prothallus of the gametophytes. Moreover, these gametophytes ceased to develop into sporophytes, indicating the essential role of myosin for gametophyte development.

Analysis of the genome of *Ceratopteris richardii* identifies a full length myosin Type XI protein sequence (CrMyoXI), with a 61.76% identity to those in seed plants. This myosin is highly expressed during the germination of spores and throughout many stages of development. In the *polka dot* mutant, where the chloroplast failed to move properly, RT PCR demonstrated that there was a decrease in the expression of CrMyoXI, suggesting that myosin may play a role in chloroplast distribution. This is the first study to show that myosin is essential for the development of gametophyte as well as organelle movement in *Ceratopteris richardii*.

RESEARCH QUESTION

How do myosins affect development of gametophytes in the fern *Ceratopteris richardii*? Do myosin in *Ceratopteris richardii* affect chloroplast distribution in prothallial cells?

HYPOTHESIS

In seed plants, the orthologue of XI has been shown to affect the myosin and has been shown to regulate cytoplasmic streaming, cell expansion, and chloroplast positioning. Although in seedless plants myosin has not been characterized, these plants possessed orthologues of myosin VIII and XI, which can be inhibited by 2,3 Butanedione monoxime (BDM). Thus, if spores of *Ceratopteris richardii* were treated with BDM, there would be a reduction in germination, cell elongation, and development since myosin is integral for these processes. Since myosin also affects the positioning of chloroplast, myosin expression may be affected in the *polka dot* mutant which showed a clumping of chloroplast in cells.

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